Neural Computing and Applications
https://doi.org/10.1007/s00521-024-10561-z

S.I.: TIMELY ADVANCES OF DEEP LEARNING WITH APPLICATIONS AND DATA DRIVEN MODELING

Novel wavelet-LSTM approach for time series prediction

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Received: 9 February 2024 / Accepted: 3 October 2024

Abstract
Time series prediction often faces challenges due to hidden patterns and noise within the data. This paper presented a novel algorithm that combines wavelet decomposition with long short-term memory (LSTM) networks, providing a distinct method for handling these challenges. The study considered the monthly rainfall data (mm) of India from January 1901 to December 2021. The series under consideration was denoised using maximum overlap discrete wavelet transform (MODWT), followed by long short-term memory (LSTM) modeling on each denoised series. The hyperband search algorithm was employed to identify the optimal hyperparameter combination for each LSTM model, aiding in further fine-tuning the model. The algorithm ended with the implementing of the inverse wavelet transform on the final predictions. In order to evaluate the efficacy of the proposed approach, it was benchmarked against the other established models such as LSTM, recurrent neural network (RNN), and artificial neural network (ANN). The result showed that the proposed model (MODWT-LSTM) significantly outperformed the other benchmark models like LSTM, RNN, and ANN in terms of forecast accuracy. Specifically, in terms of root-mean-square error (RMSE), the proposed algorithm witnessed a gain in prediction accuracy to the tune of 18.5%, 32.8%, and 36.47% than that of LSTM, RNN, and ANN model, respectively. The superiority of the proposed model is further confirmed by use of Diebold–Mariano (DM) test, establishing the hierarchy of model effectiveness as MODWT-LSTM > LSTM > RNN = ANN in terms of predictive performance.

Keywords Long short-term memory (LSTM) · Rainfall forecasting · Root-mean-square error (RMSE) · Wavelet decomposition

1 Introduction

India's diverse climate and agricultural landscape are heavily influenced by the monsoon rains, making precise rainfall prediction a vital aspect of the country's socioeconomic well-being [1]. Especially, in agriculture sector, majority of farmers are still dependent on rain because 22 out of the 36 states and union territories have irrigation coverage below the 50 percent mark for their agricultural land. Rainfed farming encompasses around 51% of the nation's net sown area which is contributing to nearly 40% of the country's overall food production. National Rainfed Area Authority (NRAA) released a key note on July 2022, in that they mentioned "climate change is likely to affect farming dependent solely on rainfall." So, rainfall prediction plays a significant role in agriculture as it allows farmers to make informed decisions about when, where, and how to grow crops and manage their resources efficiently. It is also helpful in proper management of reservoirs, floods, droughts, etc., [2].

Rainfall prediction remains challenging due to its nonlinear nature and its dependence on geographical features [3]. Traditional methods of rainfall forecasting often fall short in providing timely and precise information [4]. Many statistical models still have significant amount of processing power but the outcome of the model often fails to justify the amount of time and effort in obtaining them [5, 6]. However, in recent years, the integration of artificial

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Published online: 26 November 2024

intelligence techniques has emerged as a promising solution to enhance time series prediction. This cutting-edge approach utilizes vast datasets, advanced neural networks, and machine learning algorithms to analyze historical patterns, ultimately improving the accuracy and lead time of predictions [7]. The effectiveness of a forecast model and their output often hinges on the model's design. Therefore, a thoughtful and meticulous model design contributes significantly to achieving more accurate and precise forecasts. In this paper, we designed a novel algorithm for forecasting time series, leveraging a fusion of data-driven deep learning methodologies with wavelet decomposition techniques. Initially, the inherent noise within the dataset was addressed by employing the maximum overlap discrete wavelet transform (MODWT), a robust denoising approach. Following this preprocessing step, long short-term memory (LSTM) model was employed on each denoised series, enabling the capture of temporal dependencies of the series under consideration.

2 Background

Rainfall is indeed a highly complex metrological process, although its predictability was studied in several literature over the past years. There are two broad approaches for rainfall forecasting: First approach which involves modeling the rainfall using the fundamental physical laws governing the atmosphere is called dynamic methods [8]. The integration of the precipitable water vapor (PWV) from Global Navigation and Positioning System (GNPS) with metrological data was explored to obtain the hourly forecast of precipitation using nonlinear autoregressive neural network model (NARX) [9]. The collocated meteorological data that entail number of atmospheric parameters like temperature, PWV, pressure, relative humidity, and dew temperature were used to predict the rainfall [10]. This method is calculation intensive and also suffers from non-availability of data. Even the data are available, the complex control variables vary both in time and space. Second approach is pattern recognition method which infers about the rainfall patterns based on feature of own historical series without considering the physics of the meteorological process [11, 12]. Rainfall prediction using statistical, machine learning and deep learning models are some examples of the latter approach.

The seasonal autoregressive integrated moving average SARIMA model was utilized to forecast the rainfall in Warri Town, Nigeria [13]. Machine learning (ML) techniques emerged as an alternative to statistical techniques as it extracts the patterns and relationship between input and output without having deep understanding of physical and mathematical process [14]. The application of artificial neural network in prediction of rainfall is found in many literatures [11, 15]. A comprehensive evaluation on the predictive capabilities of Markov chain model with machine learning algorithms was started evolving [16]. In recent years, the use of deep learning algorithm for predicting rainfall have gain significant attention especially when we deal with large dataset. A novel recurrent neural network, deep echo state network (DeepESN), was employed to analyze the hourly rainfall data collected from South Taiwan [17]. The proposed model had significantly higher accuracy than SVR, back-propagation network (BPN) and ESN. The rainfall predicted using intensified long short-term memory (LSTM) model performed better than Holt–Winters, ARIMA, extreme learning machine (ELM), recurrent neural network, and LSTM [18]. Different comparisons were made between hybrid PSO-SVR, LSTM and convolutional neural network (CNN) to make 5-min and 15-min ahead rainfall forecast [19]. The result showed that PSO-SVR and LSTM were yielding similar results but significantly differ from CNN.

Preprocessing of rainfall data before applying the model leads to increase in forecast accuracy [20, 21]. They employed singular spectrum analysis along with SVR in predicting the monthly precipitation of four Chinese rainfall stations. The use of wavelet analysis helps in denoising the signal and also provides time frequency domain of the original signal, providing quality forecast [22, 23]. The wind power was predicted using discrete wavelet transform (DWT) and LSTM for short term in three wind farms of China [24]. The wavelet–SVM (WSVM) model was developed to make hourly flood forecast in the Achankovil river region of Kerala [25]. The outcome of the experiment accurately predicted the magnitude of peak discharge and peak time than SVM and wavelet–ANN (WANN) models. In this study, we are adopting maximum overlap discrete wavelet transform (MODWT), a modified version of DWT along with LSTM for forecasting the rainfall in India.

3 Research dap

While the literature covers the broader topic, there is a dearth of research on specific subtopics, such as prepocessing, optimization of hyperparameters which are summarized as follows:

1) Most of literature reviewed use DWT as a preprocessing tool for denoising the complex rainfall data, but it also suffers from drawback like fixed sample size and translation variant. To overcome this, we are adopting maximum overlap discrete wavelet transform (MODWT), a modified version of DWT for forecasting the rainfall data.

2) Many machine learning and deep learning methods are evolved to handle seasonality and nonlinearity of rainfall. However, accurate prediction can be obtained only by proper fine-tuning of hyperparameters which is still lacking.

4 Methodology

4.1 Maximum overlap discrete wavelet transform (MODWT)

The most commonly used preprocessing is the discrete wavelet analysis (DWT), which decomposes the actual signal into low and high frequency by using a predefined filter design. But, this is time variant and requires fixed sample size of length $N = 2^p$, where p is positive integer. To tackle the drawback, a modified version of DWT called maximum overlap discrete wavelet transform (MODWT) is used. Additionally, MODWT holds the benefit like elimination of down-sampling (data reduction to half), translation and time invariant DWT. Let $\{w_{j,l}\}$ be the wavelet filter and $\{s_{j,l}\}$ be the scaling filter of DWT, where j and $l = 1, \ldots, L$ represent the level of decomposition and filter length, respectively. Then, MODWT wavelet $\{\tilde{w}_{j,l}\}$ and scaling $\{\tilde{s}_{j,l}\}$ filters are defined as:

$$\tilde{w}_{j,l} = w_{j,l}/2^{\frac{j}{2}} \text{ and } \tilde{s}_{j,l} = s_{j,l}/2^{\frac{j}{2}} \tag{1}$$

The convolution of time series $X = \{X_t, t = 0, 1, 2, \ldots, N-1\}$ on the MODWT filters yields the jth level of MODWT wavelet coefficient.

$$\widetilde{D}_{j,t} = \sum_{l=0}^{L_j-1} \tilde{w}_{j,l} X_{t-l\text{modN}}, \tag{2}$$

$$\widetilde{A}_{j,t} = \sum_{l=0}^{L_j-1} \tilde{s}_{j,l} X_{t-l\text{modN}}, \tag{3}$$

where $\widetilde{D}_{j,t}$ and $\widetilde{A}_{j,t}$ refer to details and approximate part of MODWT at the j th level, respectively, and $L_j = (2^j - 1)(L-1) + 1$. The length of the MODWT wavelet coefficients at each scale will be the same as that of the original signal X, as can be seen from the preceding formulas. In matrix notation, it can be represented as follows:

$$\widetilde{D}_j = \tilde{\omega}_j X \text{ and } \widetilde{A}_j = \tilde{v}_j X \tag{4}$$

where the values of each row in $N \times N$ matrix $\tilde{\omega}_j$ are determined by $\{\tilde{w}_{j,l}\}$. Although $\{\tilde{s}_{j,l}\}$ determines the values, $\tilde{v}_j$ is given by:

$$\tilde{v}_j = \frac{1}{2^k}\begin{bmatrix} \tilde{s}_{j,0} & \tilde{s}_{j,N-1} & \tilde{s}_{j,N-2} & \tilde{s}_{j,N-3} & \cdots & \tilde{s}_{j,3} & \tilde{s}_{j,2} & \tilde{s}_{j,1} \\ \tilde{s}_{j,1} & \tilde{s}_{j,0} & \tilde{s}_{j,N-1} & \tilde{s}_{j,N-2} & \cdots & \tilde{s}_{j,4} & \tilde{s}_{j,3} & \tilde{s}_{j,2} \\ \tilde{s}_{j,2} & \tilde{s}_{j,1} & \tilde{s}_{j,0} & \tilde{s}_{j,N-1} & \cdots & \tilde{s}_{j,5} & \tilde{s}_{j,4} & \tilde{s}_{j,3} \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ \tilde{s}_{j,N-2} & \tilde{s}_{j,N-3} & \tilde{s}_{j,N-4} & \tilde{s}_{j,N-5} & \cdots & \tilde{s}_{j,1} & \tilde{s}_{j,0} & \tilde{s}_{j,N-1} \\ \tilde{s}_{j,N-1} & \tilde{s}_{j,N-2} & \tilde{s}_{j,N-3} & \tilde{s}_{j,N-4} & \cdots & \tilde{s}_{j,2} & \tilde{s}_{j,1} & \tilde{s}_{j,0} \end{bmatrix},$$

while $\tilde{\omega}_j$ is obtained by replacing $\{\tilde{s}_{j,l}\}$ by $\{\tilde{w}_{j,l}\}$. The original time series X can be retrieved using its MODWT by,

$$X = \sum_{j=1}^{J} \tilde{\omega}_j^{\mathrm{T}} \widetilde{D}_j + \tilde{v}_J^{\mathrm{T}} \widetilde{A}_J \tag{5}$$

$$X = \sum_{j=1}^{J} \widetilde{R}_j + \widetilde{S}_J \tag{6}$$

where $\widetilde{R}_j = \tilde{\omega}_j^{\mathrm{T}} \widetilde{D}_j$ and $\widetilde{S}_J = \tilde{v}_J^{\mathrm{T}} \widetilde{A}_J$ refers to details and approximate part of MODWT at the j th level and J level, respectively.

4.2 Long short-term memory (LSTM)

Time series prediction works on the principle that current values depend on the past observations. Unlike traditional ANNs that process inputs independently, RNNs are designed to retain information from previous time steps, making them well-suited for tasks involving sequences or time series data. RNNs have a feedback connection that allows them to retain information from previous steps in the sequence and use it to make predictions or make decisions at each step. The primary shortcoming of RNNs lies in their ability to handle long-term dependencies. While it is theoretically feasible to manage extended contextual relationships within data, RNNs often struggle to effectively learn and utilize them in real-world applications.

LSTM, a special kind of RNN, is promising when dealing with large and complex datasets, and when there is a need to capture intricate temporal and spatial dependencies in time series. In RNNs, the repetitive module consists of a straightforward tanh function, while in LSTMs, the repeating module has more intricate and complex structure which are explained as follows:

Cell state: The cell state flows across the network, with information being added or removed through the use of gating mechanisms. An LSTM cell contains three gates: a forget gate, an input gate, and an output gate.

Forget gate: It includes a sigmoid layer that determines which information should be removed from the cell state. This is obtained by combining the current input x_t with the output of the LSTM unit h_{t-1} from last iteration.

$$f_t = \sigma\left(W^f.x_t + U^f.h_{t-1} + b^f\right) \quad (7)$$

where W^f and U^f are the weight associated with x_t and h_{t-1}, respectively, while b^f represent bias vector for forget gate.

Input gate: It comprises two layers: The initial layer is a sigmoid layer, responsible for making decisions about updating the values, while the subsequent layer is a tanh layer, which produces a vector of new candidate values, denoted as $\widetilde{C}_t$, that can be incorporated into the cell state.

$$i_t = \sigma\left(W^i.x_t + U^i.h_{t-1} + b^i\right) \quad (8)$$

$$\widetilde{C}_t = tanh(W^c.x_t + U^c.h_{t-1} + b^c) \quad (9)$$

where W^i and U^i are the weight associated with x_t and h_{t-1}, respectively, while b^i represents bias vector for input gate. Then, we need to update the old cell state C_{t-1} into new state C_t. Update include forgetting the old information by multiply the f_t with C_{t-1} and add the new info by multiply i_t with $\widetilde{C}_t$.

$$C_t = \left(f_t * C_{t-1} + i_t * \widetilde{C}_t\right) \quad (10)$$

Output gate: The output gate determines the value of the upcoming hidden state h_t, which encapsulates information from prior inputs. Initially, both the current state and the preceding hidden state values are input into the sigmoid function. Subsequently, the new cell state C_t derived from the existing cell state undergoes processing through the tanh function.

$$O_t = \sigma(W^o.x_t + U^o.h_{t-1} + b^o) \quad (11)$$

$$h_t = O_t * \tanh(C_t) \quad (12)$$

where W^o and U^o are the weight associated with x_t and h_{t-1}, respectively, while b^o represents bias vector for output gate. Figure 1 presents the unit of LSTM cell.

4.3 Proposed MODWT-LSTM algorithm

In Fig. 2, the proposed algorithm combines MODWT with LSTM for time series prediction. MODWT offers advantages over traditional preprocessing methods, such as discrete wavelet transform (DWT), by providing a more flexible and efficient denoising technique. Additionally, the study focuses on optimizing hyperparameters to fine-tune the LSTM model, which further helps in fine-tuning the model. The culmination of proposed algorithm involves applying the inverse wavelet transform to the final predictions, ensuring the restoration of the original data format. The implementation of the MODWT-LSTM model involves a systematic approach that combines wavelet decomposition and LSTM modeling as presented below:

Step 1: Decompose the data using the Haar filter, i.e., from 6 to 12 levels, and find the optimal level of decomposition (J).

$$X = \widetilde{D}_1 + \widetilde{D}_2 + \ldots + \widetilde{D}_J + \widetilde{A}_J \quad (13)$$

Step 2: Define the training testing ratio of the given data (e.g., 90:10).

Step 3: For each decomposed series (i.e., $\widetilde{D}_\alpha where \alpha = 1,2,\ldots J$) retrieved in Step 1.

- Normalize the decomposed series using Minmax scaler

$$D'_\alpha = \text{Min}\max\left(\tilde{D}_\alpha\right) \quad (14)$$

- Split the dataset into training and validation subsets based on the defined training ratio.

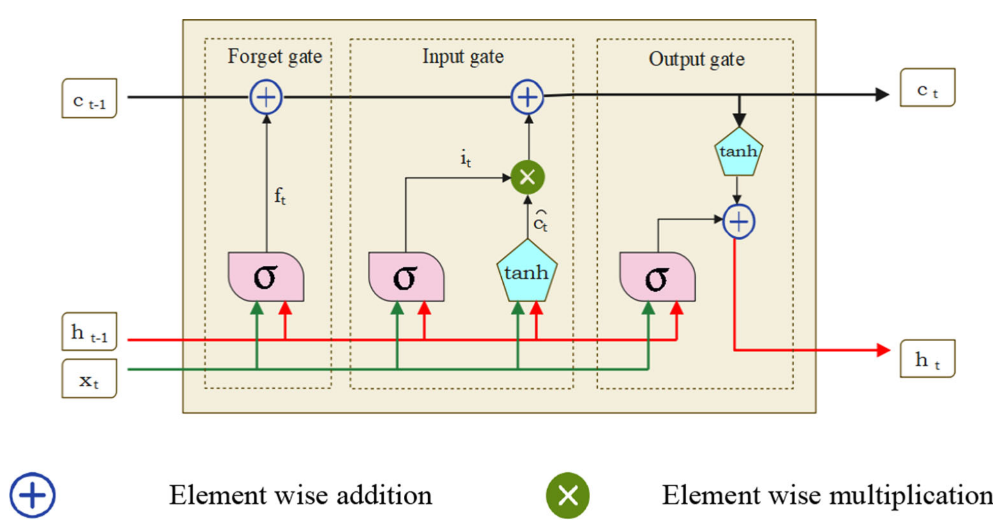

Fig. 1 The unit of LSTM cell

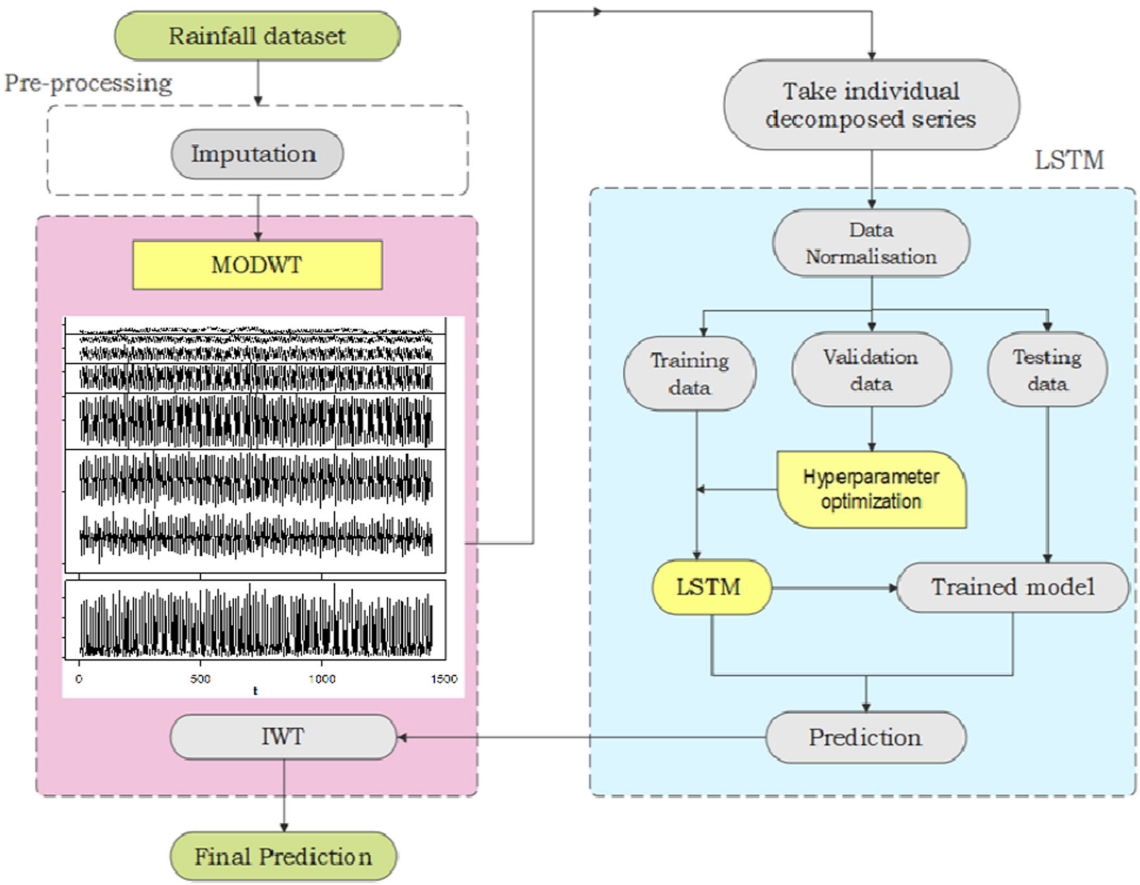

Fig. 2 Step involved in MODWT-LSTM model

- Train an LSTM model on the training data by defining the LSTM model architecture (e.g., input, hidden layers) and obtain the optimum hyperparameters, and then Eqs. 7–12 become,

$$f_{D'_{\alpha t}} = \sigma\left(W^f_\alpha . D'_{\alpha t} + U^f_\alpha . h_{\alpha(t-1)} + b^f_\alpha\right) \tag{15}$$

$$i_{D'_{\alpha t}} = \sigma\left(W^i_\alpha . D'_{\alpha t} + U^i_\alpha . h_{\alpha(t-1)} + b^i_\alpha\right) \tag{16}$$

Run the LSTM model on test data using optimized hyperparameter and obtain prediction from each series.

$$O_{D'_{\alpha t}} = \sigma\left(W^o_\alpha . D'_{\alpha t} + U^o_\alpha . h_{\alpha(t-1)} + b^o_\alpha\right) \tag{17}$$

where $f_{D'_{\alpha t}}, i_{D'_{\alpha t}} and O_{D'_{\alpha t}}$ represent the output of the normalized decomposed series at α $^{\text{th}}$ level regarding the forget gate, input gate and output gate, respectively.

Substituting (4) in (17)

$$O_{D'_{\alpha t}} = \sigma\left(W^o_\alpha . \left(\tilde{\omega}'_j X\right) + U^o_\alpha . h_{\alpha(t-1)} + b^o_\alpha\right) \tag{18}$$

$$\widehat{O_{D'_{\alpha t}}} = inv \min\max\left(O_{D'_{\alpha t}}\right) \tag{19}$$

Substituting (19) in (18)

$$\widehat{O_{D'_{\alpha t}}} = inv \min\max\left(\sigma\left(W^o_\alpha . (\tilde{\omega}'_j X) + U^o_\alpha . h_{\alpha(t-1)} + b^o_\alpha\right)\right) \tag{20}$$

Step 4: Perform wavelet reconstruction using inverse wavelet transform (IWT) on all predicted series obtained in Step 3 to get the final prediction.

$$\text{Final prediction } = \sum_{\alpha=1}^{J} \widehat{O_{D'_{\alpha t}}} + \widehat{O_{A'_{Jt}}}$$

From all the above equations,

$$\begin{aligned}\text{Final prediction}\\ &= \sum_{\alpha=1}^{J} \text{inv}\min\max\left(\sigma\left(W^o_\alpha . \left(\tilde{\omega}'_j X\right) + U^o_\alpha . h_{\alpha(t-1)} + b^o_\alpha\right)\right)\\ &\quad + \text{inv}\min\max\left(\sigma\left(W^o_J . \left(\tilde{v}'_j X\right) + U^o_J . h_{\alpha(t-1)} + b^o_J\right)\right)\end{aligned}$$

5 Empirical study

5.1 Data description

The study considered monthly rainfall (mm) of India from January 1901 to December 2021. The dataset contains 1452 data points and has been prechecked to detect and manage missing values and duplicates. The data are split into a ratio of 90% for training, using 1309 data points to build the model, and 10% for testing, where 143 data points are reserved to validate the model. Figure 3 depicts monthly rainfall from Jan 1901 to Dec 2021.The average of total annual rainfall in India over the years (1901 to 2021) is approximately1176 mm. The annual precipitation ranges from a minimum of 920 mm in the year 2002 to a maximum of 1464 mm in the year 1917. The percentage contribution of each season to the total annual rainfall is as follows: The winter season (January–February) contributes approximately 3.53% of the total annual rainfall. The pre-monsoon season (March–May) accounts for around 10.91% of the total annual rainfall. The Monsoon Season (June-Sept) is the dominant contributor, contributing approximately 75.20% of the total annual rainfall. The post-monsoon season (Oct-Dec) contributes approximately 10.27% of the total annual rainfall. These percentages provide a clear breakdown of how each season contributes to the overall annual precipitation. The descriptive statistics of monthly rainfall is presented in Table 1.

The data show a wide range of monthly rainfall values with a positive skewness (1.01) and platykurtic (-0.30) indicating that higher values are more prevalent and has lighter tails than a normal distribution, respectively. The p-value of both Jarque–Bera test and Shapiro–Wilk's test is less than 0.01, confirming that the data may not follow a normal distribution.

Table 1 Descriptive statistics

Descriptive statistics	Monthly rainfall (mm)
Mean	97
Minimum	1.6
Maximum	375.5
Standard Deviation	96.98
Kurtosis	− 0.30
Skewness	1.01
C.V. (%)	98.97
Jarque–Bera test	0.82 (< 0.01)
Shapiro–Wilk's test	252.39 (< 0.01)

*The values in the parentheses indicate p-value

5.2 Selection of level of decomposition of wavelet

Haar wavelet is used in the study to decompose the original series. The better time (spatial) localization and computational efficiency of Haar transforms make them valuable in many applications [26]. The precision of the features extracted from a time series is contingent on the chosen level of decomposition. Consequently, the selection of a suitable decomposition level is a crucial task in the design of analysis. The minimum and maximum level of decomposition is denoted by $L_{\min}$, $L_{\max}$ [25] and given as,

$$L_{\min} = \text{int}\ (\log N)$$

$$L_{\max} = int(\log_2 N)$$

where N is the number of observations.

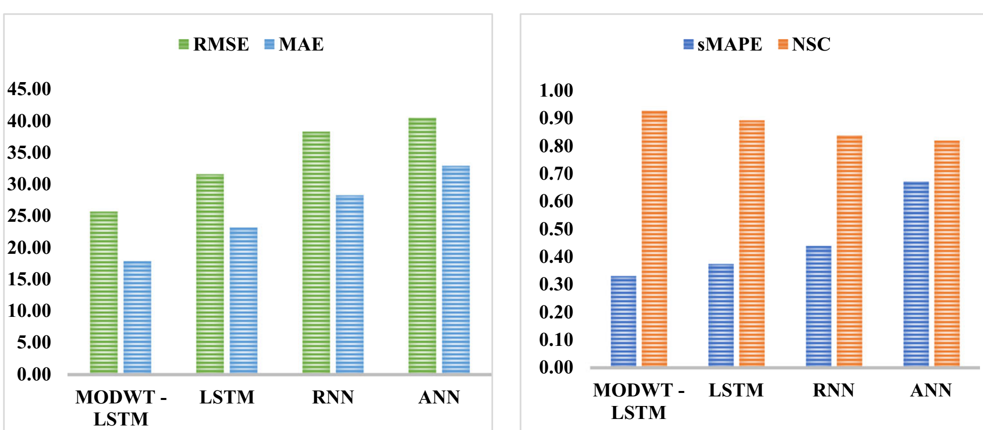

Fig. 3 Accuracy measure of different model for testing data

5.3 Implementation of MODWT- LSTM model

After decomposition, each decomposed series is fed into separate LSTM model for training the data. Deep learning typically demands significant time and computational power during the training process, motivating the quest for efficient optimization algorithms. While distributed parallel training can enhance deep learning speed, it doesn't substantially reduce the computational resources required. Thus, only optimization algorithms that are resource-efficient and promote faster model convergence can fundamentally enhance learning speed and enhance the forecasting capabilities of neural networks. There are different search algorithms like Bayesian optimization, hyperband, random search, etc., exist to discover the optimal hyperparameter combination. The Hyperband search algorithm examines a greater number of hyperparameter combinations within a predefined resource allocation during each iteration. This resource allocation can involve factors such as training epochs, datasets, training duration, or similar resources that influence the training process. So, this was employed in this context by using Keras -tuner, and the exploration of the hyperparameter space is detailed in Table 2.

6 Result and discussion

The monthly rainfall time series data were subjected to MODWT decomposition, where the minimum and maximum level of decomposition was 3 and 7, respectively. The optimal level of decomposition was found to be 6 by graphical way [27]. Subsequently, every decomposed series was considered as input into LSTM to train the models, and the best hyperparameters were acquired for each individual series. The length of the input data was fixed in LSTM as 24. The optimum hyperparameter for different decomposed series is exhibited in Table 3. After the models have undergone training, the test data are employed to validate and assess the performance of the models. This process helps ensure that the models generalize well and make more precise predictions when presented with new, unseen data.

For the comparison purpose, the actual series was fitted with other models, including LSTM, RNN and ANN. The evaluation criteria used for comparison are root-mean-square error (RMSE), mean absolute error (MAE), symmetric mean absolute percentage error and Nash–Sutcliffe coefficient (NSC). There are defined as follows:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(A_i - \hat{P}_i\right)^2}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|A_i - \hat{P}_i\right|$$

$$sMAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|A_i - \hat{P}_i\right|}{\left(|A_i| - |\hat{P}_i|\right)/2}$$

$$NSC = 1 - \frac{\sum_{i=1}^{N}\left(A_i - \hat{P}_i\right)^2}{\sum_{i=1}^{N}\left(A_i = \overline{A}\right)^2}$$

where A_i and $\hat{P}_i$ represent the actual and predicted value of data, respectively, and N is the total number of observations. Training data is employed to build and train the deep learning model. During this phase, the model learns patterns, relationships, and features in the data, allowing it to make predictions. After training, the model needs to be evaluated to ensure it can generalize its predictions to new, unseen data called test data. Table 4 presents the result of MODWT-LSTM model with other models in terms of four evaluation criteria for train and test data. Lower values for RMSE, MAE, and sMAPE are indicative of better predictive accuracy, while a higher NSC value suggests a better information capture.

In both training and testing dataset, the MODWT-LSTM model's excellence over other models was confirmed by the lowest RMSE, MAE, and sMAPE values, which collectively indicated exceptional performance. This assertion was further supported by the fact that the MODWT-LSTM model obtained the highest NSC value, underscoring its excellence. Figure 3 displays accuracy measures of different models for test data. Particularly, in test data, the RMSE of MODWT-LSTM model was 18.5%, 32.8%, and 36.47% lower than that of LSTM, RNN, and ANN, respectively. MODWT-LSTM had the lowest MAE value (17.8) depicting the best performance of the model. It also has the lowest sMAPE (0.3309) on the testing set, indicating its ability to maintain accuracy when applied to unseen data. Again, MODWT—LSTM has a relatively

Table 2 Different hyperparametric search for the LSTM model

Hyperparameters	Search space
No of LSTM units	Min value = 16, Max value = 256 at the step rate of 8
No of epochs	Min value = 10, Max value = 100 at the step rate of 10
Batch size	16, 32, 64, 128
Learning Rate	0.01, 0.001, 0.0001

Table 3 Optimized results for each decomposed series

Haar filter (level 6)	Optimum hyperparameters			
	No of LSTM units	No of epoch	Batch size	Learning rate
D1	16	10	32	0.0001
D2	80	20	64	0.0001
D3	176	50	128	0.0001
D4	48	20	64	0.001
D5	208	30	16	0.001
D6	208	40	16	0.0001
A1	64	70	16	0.01

Table 4 Evaluation Criteria for the different models for training and testing dataset

Price series	Models	RMSE	MAE	sMAPE	NSC
Training set	MODWT—LSTM	**26.1760**	**18.6166**	**0.2984**	**0.9277**
	LSTM	33.7715	25.6545	0.6177	0.8796
	RNN	36.3151	28.7037	0.5952	0.8608
	ANN	38.8125	29.2685	0.5302	0.8410
Testing set	MODWT—LSTM	**25.6847**	**17.8884**	**0.3309**	**0.9267**
	LSTM	31.5729	23.1936	0.3750	0.8916
	RNN	38.2876	28.2482	0.4394	0.8373
	ANN	40.4715	32.8906	0.6711	0.8182

high NSC (0.9267), which suggests that it likely excels in capturing the underlying information when compared to the other models. The plot of actual and predicted value (MODWT–LSTM model) is presented in Fig. 4.

The Diebold–Mariano (DM) test is a statistical evaluation method employed to assess and compare the predictive accuracy of two or more forecasting models. The results showed that the MODWT-LSTM significantly differs from LSTM, RNN and ANN models. Similarly, the LSTM also significantly differs from RNN and ANN. But, RNN was having same accuracy as that of ANN. In terms of predictive performance on this dataset, the ranking of model effectiveness can be expressed as MODWT—LSTM > LSTM > RNN = ANN. The result given in Table 5 clearly highlights significant differences in performance between various model pairs, underscoring the effectiveness of the MODWT-LSTM approach compared to traditional LSTM,

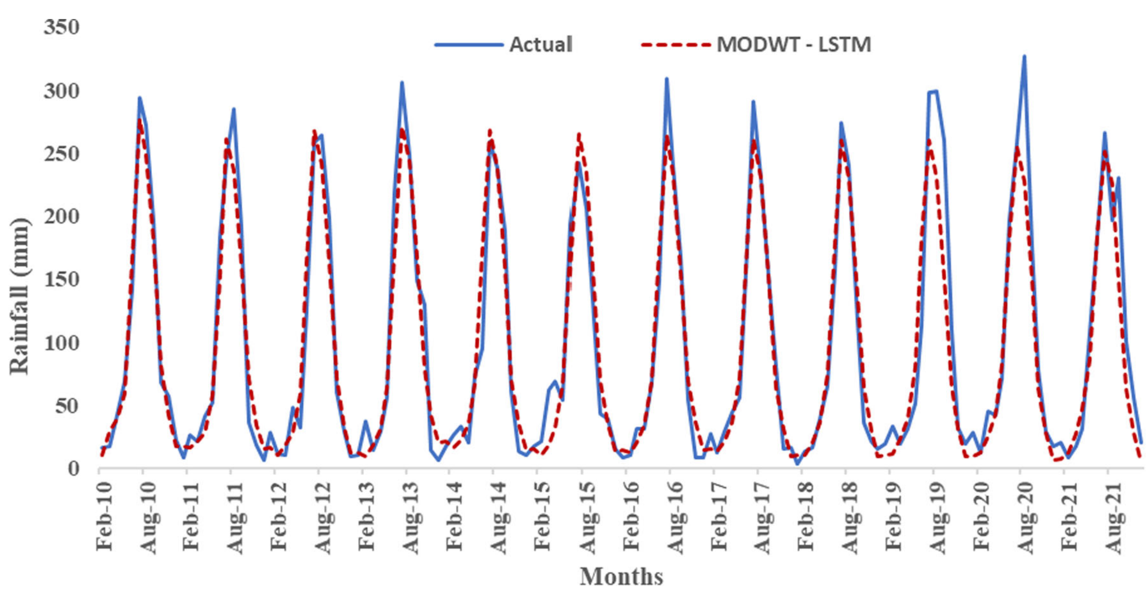

Fig. 4 Plot of actual and predicted values for MODWT-LSTM model on test data

Table 5 DM test results for comparison of different models

Price series	Models	MODWT—LSTM	LSTM	RNN	ANN
Testing set	MODWT—LSTM	–	− 2.2315 (< 0.01)	− 3.5382 (< 0.01)	− 5.3334 (< 0.01)
	LSTM		–	− 3.9348 (< 0.01)	− 5.6662 (< 0.01)
	RNN			–	− 1.0239 (> 0.01)
	ANN				–

*The values in the parentheses indicate p-value

RNN, and ANN models for the given dataset and prediction task.

7 Conclusion

Rainfall prediction is currently and will continue to be a prominent research focus for researchers because of its interdependence with daily life. In this study, we used the wavelet decomposition to denoise the series under consideration followed by deep learning model for precise prediction of rainfall. One of the most significant limitations encountered when using wavelet decomposition in predictive modeling is the choice of the decomposition level. If there are too many decomposition levels, the model can lose important non-noise information, leading to data distortion. Conversely, using too few decomposition levels leaves a substantial amount of noise in the data, adversely affecting model accuracy. This challenge was addressed meticulously by selecting a decomposition level of six as there was tight frequency band observed in the trend of the data under consideration. By opting for this intermediate decomposition level, we aimed to strike a balance between preserving essential information and mitigating noise interference, thereby optimizing the predictive performance of our model. Following the decomposition process, the detailed and approximate components are considered input into distinct LSTM models. Since, rainfall exhibits seasonal variations and LSTMs can automatically adapt to these patterns, making them well-suited for capturing both short-term and long-term trends in rainfall data. Selecting suitable hyperparameters for each series poses another challenge, but this difficulty was addressed by utilizing the Hyperband search algorithm. It efficiently explores the hyperparameter space, fine-tuning the model to achieve optimal performance. The prediction obtained from each series were combined using inverse wavelet transform. The proposed model (MODWT-LSTM) was compared with other models for comparing the prediction efficiency. The results indicated a significant reduction in forecast errors for the proposed MODWT-LSTM model compared to benchmark models such as LSTM, RNN, and ANN. Specifically, in terms of root-mean-square error (RMSE), the proposed algorithm witnessed a gain in prediction accuracy to the tune of 18.5%, 32.8%, and 36.47% than that of LSTM, RNN, and ANN model, respectively. The superiority of the proposed model is further confirmed by the results of the DM test as MODWT-LSTM model significantly outperformed the other benchmark models. This model exclusively rely on historical rainfall data rather than broader climatic patterns and external factors affecting rainfall. However, its narrow focus allows for versatile application across diverse regions and situations, enhancing its generalization. By accurately forecasting rainfall based on historical trends, it aids authorities and organizations in preparing for and responding to natural disasters such as floods or droughts. Furthermore, it offers valuable insights into long-term precipitation patterns, supporting water resource management and agricultural planning efforts.

Acknowledgements The authors are thankful to the anonymous reviewers for their suggestions.

Funding This research received no external funding.

Data availability Data will be available on reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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